

Abstract No.

Cellular Migratory Scaling Stratifies Glioblastoma Survival: PTEN Amino Acid Substitutions as Pro-Invasive Drivers

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Glioblastoma (GB) infiltration is traditionally quantified by kinetic metrics such as speed or distance, which often fail to capture the strategic nature of tumor spreading. In this study, we hypothesized that "migratory efficiency", the mathematical ability of a cell to optimize environmental exploration, serves as a superior predictor of aggressiveness and reflects a distinct molecular landscape. Using cell live imaging on 30 patient-derived GB primary cultures (IDH-WT) to track 4,137 individual trajectories, we applied Diffusion Entropy Analysis (DEA) to calculate the Delta Scaling parameter (d), a scale-invariant metric of motion. This framework allowed us to stratify tumors into Low (L), Medium (M), and High (H) motility groups based on their exploratory efficiency. Our results demonstrate that high d scaling (H) identifies a "highly efficient" phenotype characterized by structured, super-diffusive movement, which directly correlates with significantly shorter patient survival. Multi-omic analysis revealed that Group H is uniquely enriched in PTEN mutations (75% vs. 8% in Group L). Specifically, PTEN amino-acid substitution (AA) carriers in an expanded cohort (n=50) exhibited a dismal median survival of 6.7 months, significantly shorter than both PTEN wild-type cases (16.6 months, p=0.01) and those with canonical truncating (Ter) mutations (14.7 months, p=0.03). Notably, while complete PTEN loss (Ter) showed no significant survival difference from wild-type, the presence of specific AA substitutions significantly worsened the clinical prognosis. Our study demonstrates that migratory efficiency is a powerful predictor of GB prognosis. We provide evidence for a critical paradigm shift: while PTEN loss is a genomic hallmark, specific residual missense mutations correlate with a highly efficient invasive program. This suggests that in the context of habitual gene loss, these alterations may act as pro-invasive or oncogenic drivers, worsening the clinical scenario far beyond the effects of canonical PTEN truncation, offering novel, actionable therapeutic targets for glioblastoma and potentially other PTEN-deficient cancers.



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